CyberGuard: Cybersecurity Awareness Assistant

Project Overview

CyberGuard is a sophisticated virtual assistant designed to educate South African citizens on cybersecurity topics in a conversational manner. As cyber threats rise significantly in South Africa, this tool serves as an interactive educational platform to help users identify and mitigate risks such as phishing, malware, and social engineering.

This project is part of the **PROG6221 (Programming 2A)** Portfolio of Evidence (POE).

Features

- **Interactive Chat Interface**: A modern, dark-themed WPF GUI for engaging user interaction.
- **Comprehensive Knowledge Base**: Detailed information on 9+ cybersecurity topics including VPNs, Phishing, Passwords, MFA, and more.
- **Educational Guidance**: Provides specific "Tips" and "Prevention" strategies for each topic.
- **Speech Synthesis**: Integrated text-to-speech capabilities with mute/unmute functionality.
- **Session Reporting**: Generates a full, timestamped chat history report for user review.
- **Input Validation**: Robust name validation and sanitized chat inputs.

Cybersecurity Topics Covered

- **VPN**: Masking IP addresses and securing public Wi-Fi.
- **Phishing**: Identifying deceptive emails and links.
- **Passwords**: Creating strong passphrases and using managers.
- **MFA**: Implementing multi-factor authentication.
- **Malware**: Protecting against viruses and ransomware.
- **Firewalls**: Monitoring and controlling network traffic.
- **Social Engineering**: Recognizing human manipulation tactics.

- **Privacy**: Managing online data and app permissions.
- **General Tips**: Core safety practices for everyday digital life.

Getting Started

Prerequisites

- .NET 6.0 SDK or higher
- Windows OS (for WPF support)
- `System.Speech` library (included in Windows)

Installation & Running

1. **Clone the repository**:

   ```bash
   git clone https://github.com/USER_NAME/CybersecurityChatBot_2.git
   ```

2. **Navigate to the project directory**:

   ```bash
   cd CybersecurityChatBot_2
   ```

3. **Build the project**:

   ```bash
   dotnet build
   ```

4. **Run the application**:

   ```bash
   dotnet run
   ```

GitHub Requirements Compliance

- **Commit History**: Minimum of 6 commits per module (Total 36+ commits) detailing the incremental development.

- **Continuous Integration (CI)**: GitHub Actions implemented via `.github/workflows/dotnet.yml`.
- **Documentation**: Comprehensive README with clear setup and usage instructions.

License

This project is for educational purposes as part of the IIE PROG6221 module.

References

Pieterse, H. 2021. The Cyber Threat Landscape in South Africa: A 10-Year Review. *The African Journal of Information and Communication*, 28(28). doi: https://doi.org/10.23962/10539/32213